Homework 10

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1 H 10-1

This is the same project from the worksheet, so we have a 50% chance of a cashflow of 140 and a 50% chance of a cashflow of 90, which means that after discounting at a weighted average cost of capital of 15% our cashflow has a present value of $\frac{0.5\times(140+90)}{1.15} = 100$. We assume the entrepreneur raises 70 in debt, so the equityholder's cashflow is $0.5 \times 0.7 \times (140 + 90) = 80.5$ and they need to hold $\frac{80.5}{1.15} = 70\%$ of the company to be adequately compensated for their risk.

2 H 10-2

We are given that the debt-to-equity ratio of 0.4, an equity beta of 1.4, and a debt beta of 0.2. We can interpret the first as $\frac{D}{E} = 0.4$, which means all further ratios will be satisfied if we write $D = 0.4$ and $E = 1$. That lets us solve for the unlevered equity beta:

$$\beta_U = \frac{E}{D+E}\beta_E + \frac{D}{D+E}\beta_D = \frac{1}{1.4} \times 1.4 + \frac{0.4}{1.4} \times 0.2 = 1 + \frac{0.4}{7} = \frac{1.4}{7} = 0.2$$

Now we hold this fixed and reverse-engineer the equity beta of 1.2:

$$0.2 = \frac{1}{1.4} \times 1.2 + \frac{0.4}{1.4}\beta_D \implies \beta_D = \frac{7}{2}\left(0.2 - \frac{5}{7}\right) = \frac{7}{10} - \frac{5}{2} = 0.7 - 2.5 = -1.8$$

3 H 10-3

The weighted average cost of capital is given by

$$r_U = \frac{E}{D+E-C} r_E + \frac{D}{D+E-C} r_D - \frac{C}{D+E-C} r_f = \frac{20}{20+40-4}(0.04 + 1.2 \times 0.06) + \frac{40}{20+40-4}(0.04 + 0.1 \times$$

If we instead use the cash to clear debt and hold the betas constant, we instead can find an unlevered equity beta so that we have a beta for the asset:

$$\beta_U = \frac{40}{40+16} \times 1.2 + \frac{16}{40+16} \times 0.1$$